



Dart Hands-On Coding Lab Assignments

Instructions: Copy these starter snippets into your code editor (like VS Code or DartPad) and complete the tasks.



Part 1: String & Number Methods Tasks

1. Task 1: Clean and Convert User Input

A user accidentally typed their favorite language with extra spaces: " dart programming ". Write code to clean the outer empty spaces and change the letters to all uppercase.

```
void main() {  
  String text = "  dart programming  ";  
  
  // 1. Write your code here to print the cleaned uppercase text  
  
}
```

2. Task 2: Slicing sentences into Lists

Take the string below and chop it up everywhere a comma , appears so that it prints as a clean, structured List container.

```
void main() {  
  String items = "apple,banana,orange,grape";  
  
  // 2. Write your code here to split and print the collection  
  
}
```

3. Task 3: Correcting Negative Bank Ledger Entries

A banking account shows a balance value of -85.64. Write code using built-in methods to drop the negative sign (convert to absolute value) and round it to the nearest whole integer number.

```
void main() {  
  double accountBalance = -85.64;  
  
  // 3. Write code to turn it positive, round it, and print the output  
  
}
```



Part 2: Complex Collections & Data Structures Access Tasks

1. Task 4: Extracting data from a List inside a Map

Given the following restaurantOrder map configuration layout, write a print statement that uses indexing brackets to extract and display only the second item ("**Mushrooms**") from the toppings list.

```
void main() {
    Map<String, dynamic> restaurantOrder = {
        "customerName": "Sophia",
        "table": 5,
        "toppings": ["Extra Cheese", "Mushrooms", "Black Olives"]
    };

    // 4. Print "Mushrooms" by accessing the map key and list index position
}
```

2. Task 5: Navigating and Updating a Deeply Nested Map

Given the following multi-layered deviceProfile configuration dataset, write code to:

A) Print out the city value ("**Kathmandu**") by chaining map keys.

B) Update the nested storage parameter value from "128GB" to "256GB".

```
void main() {
    Map<String, dynamic> deviceProfile = {
        "owner": "Rohan",
        "device": "Laptop",
        "specs": {
            "storage": "128GB",
            "location": {
                "city": "Kathmandu",
                "country": "Nepal"
            }
        }
    };

    // 5A. Write a print statement to display the city name

    // 5B. Write code to update the storage parameter to "256GB" and print the updated map
}
```